

DEVA ANAND

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Education

MS, Computer Science, University of Massachusetts Amherst 2025–2027
Advanced NLP, Advanced Machine Learning, Optimization in CS, Systems for Data Science, Performance Engineering

BE, Computer Science and Engineering, Vels Institute of Science, Technology and Advanced Studies 2019–2023

Professional Experience

Wysa, Backend Engineer | *Bengaluru, India* Jul 2023 – Jul 2025

- Owned backend security across a multi-tenant NHS eTriage platform: **AWS KMS** encryption for PII/clinical data, authentication hardening, rate limiting, audit logging, and remediation of **20+ VAPT findings**.
- Engineered the multi-tenant eTriage backend (Node.js, MongoDB) for **8+ UK clinical clients**, processing **10,000+ monthly submissions** via Mayden iaptus REST integrations; contributed to **\$340K+** in revenue.
- Architected a full-stack annotation platform (React frontend, Node.js/MongoDB backend) used daily by the AI team, cutting manual work by **100+ hours/month** and lifting throughput by **60%**.
- Served as primary technical POC in an Agile/Scrum team, adding integration tests for regression protection, mentoring a second developer to on-call readiness, and piloting a COO-sponsored GitHub Copilot experiment.

Cario Growth Services, Backend Developer Intern | *Chennai, India* Nov 2022 – May 2023

- Designed and shipped **20+ RESTful APIs** (Node.js, Fastify, PostgreSQL) with object-oriented service design, input validation, clear contracts, and backward-compatible changes used in production.
- Migrated core repositories from JavaScript to TypeScript and shipped a production LLM auto-commenting feature (**4-bit quantized Llama-7B via llama.cpp**) with async queued generation and content-safety filtering.

Freelance / Open Source, Full-Stack Engineer | *Remote* Jul 2022 – Aug 2023

- Built and deployed **Underdogs Fitness** (Website), a production MERN platform (React) with **JWT-based authentication**, Stripe hybrid payments, and cron automation; live **3+ years**.

Projects

AgenticSearch – Provenance-First Entity Discovery (GitHub | Demo) Python, FastAPI, OpenAI / Groq, SQLite

- Built a Python/FastAPI backend service turning open-ended queries into structured entity tables via typed retrieval planning, reranking, deterministic extraction, and output verification with cell-level provenance; hardened with **150+ automated tests** and provider routing across OpenAI/Groq, deployed as a cloud service on Google Cloud Run.

cf-ai-research-scout – Edge-Native Cloud Service (GitHub) TypeScript, Cloudflare Workers, Durable Objects, D1

- Designed and built a scalable cloud service in **TypeScript** with durable, restart-safe workflows and D1 storage, streaming an AI agent over WebSocket that survives process restarts and individual tool failures.

Sonare – Offline Sign / Speech App (GitHub) React, Electron, FastAPI, MediaPipe, whisper.cpp

- Engineered a **React**/Electron frontend and FastAPI backend integrating camera, microphone, and local services (MediaPipe hand tracking, whisper.cpp ASR, local TTS) into one offline application, with async APIs and queues keeping the UI responsive under local video and audio streams.

Publications

- “Alzheimer’s Disease Classification using Transfer Learning,” IEEE CONIT 2023, first author, deep transfer learning on neuroimaging data. (Paper)

Skills

Languages & Databases: Java, Python, TypeScript, JavaScript, SQL, MySQL, PostgreSQL, MongoDB, Redis

Backend & APIs: FastAPI, Node.js, Express, REST APIs, object-oriented design, async services, multi-tenant architecture

Testing & CI: pytest, integration testing, LambdaTest, Git, GitHub Actions, CI/CD, Agile / Scrum

Security: AWS KMS encryption, authentication, JWT, rate limiting, audit logging, VAPT remediation

Cloud & Platforms: AWS, Azure, GCP, Linux, Docker, MongoDB Atlas

Achievements

- Selected for **GirlScript Summer of Code 2023**; contributed to Linkfree, Freehit, and ProjectsHut. - Tutored **30+ underprivileged students** in programming at Sayur, a non-profit in Tamil Nadu.